

Program 2

```

ZZJOB 5                                CSC 001   GERALD, C. F.
ZZFORX5
*LIST PRINTER
C PROGRAM FOR ROMBERG INTEGRATION
C PROGRAM BEGINS WITH TRAPEZOIDAL INTEGRATION WITH 10 SUBINTERVALS
C INTERVALS ARE HALVED AND RESULTS ARE EXTRAPOLATED UP TO EIGHTH ORDER
C OF EXTRAPOLATION.
C MAXIMUM NUMBER OF SUBINTERVALS USED IN PROGRAM IS 2560.
C DEFINE INTEGRAND WITH AN ARITHMETIC STATEMENT FUNCTION.
      FCN(X) = SIN(X)/X
      DIMENSION TRAP(9,9)
C READ UPPER LIMIT AND TOLERANCE AND COMPUTE FIRST INTEGRAL.
      READ 100, B, TOL
      H = B/10.
      SUM = 1. + FCN(B)
      X = 0.0
      DO 10 I = 2,10
        X = X + H
      10 SUM = SUM + FCN(X)*2.
      TRAP(1,1) = H/2.*SUM
C RECOMPUTE INTEGRAL WITH H HALVED, EXTRAPOLATE AND TEST. REPEAT UP TO 8 TIMES
      DO 20 I = 1,8
        H = H/2.
        X = H
        K = 10.**2.**I
        DO 30 J = 2,K*2
          SUM = SUM + FCN(X)*2.
        30 X = X + H + H
          TRAP(I,I+1) = H/2.*SUM
          DO 40 II = 1,I
            TRAP(II+1,I+1) = TRAP(II,I+1) + 1./(4.**II - 1.)*(TRAP(II,I+1) -
              1 TRAP(II,I))
            IF (ABSF(TRAP(I+1,I+1) - TRAP(I,I+1)) - TOL) 50,50,20
        20 CONTINUE
C IF TOLERANCE NOT MET AFTER 8 EXTRAPOLATIONS, PRINT CALCULATED VALUES AND NOTE
      PRINT 200, B
      DO 60 K = 1,9
        60 PRINT 203, (TRAP(J,K), J = 1,K)
          PRINT 201, TRAP(9,9), TOL
          CALL EXIT
C PRINT RESULTS WHEN TOLERANCE IS MET.
      50 PRINT 200, B
        I = I + 1
        DO 70 L = 1,I
          70 PRINT 203, (TRAP(J,L), J = 1,L)
            PRINT 202, TRAP(1,1), K, TOL
            CALL EXIT
      100 FORMAT (F10.0, E10.0)
      200 FORMAT (// 38H VALUE OF SINE INTEGRAL BETWEEN 0 AND , F6.2,
        1 24H BY ROMBERG INTEGRATION / )
      201 FORMAT (/ 10H VALUE IS , F10.7, 31H BUT DID NOT MEET TOLERANCE OF
        1 E14.7)
      202 FORMAT (/ 10H VALUE IS , F10.7, 2H, / 15,
        1 49H SUBINTERVALS WERE REQUIRED TO MEET TOLERANCE OF , E14.7)
      203 FORMAT (1H ,8F10.7)
      END
15.0      2.E-7
ZZZZ

```

VALUE OF SINE INTEGRAL BETWEEN 0 AND 15.00 BY ROMBERG INTEGRATION

```

1.6077255
1.6156591 1.6183036
1.6175654 1.6182008 1.6181940
1.6180369 1.6181940 1.6181936 1.6181936

```

VALUE IS 1.6181936.

80 SUBINTERVALS WERE REQUIRED TO MEET TOLERANCE OF 2.0000000E-07